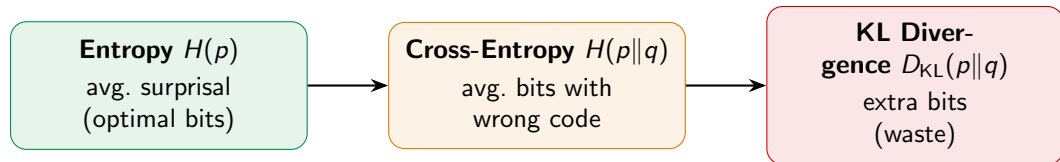


Information Theory II: ML Connections

KL = MLE · Cross-Entropy Loss · Forward/Reverse KL · Mutual Information

Recap: The Three Pillars



$$H(p||q) = H(p) + D_{KL}(p||q)$$

Today: We connect these concepts to machine learning.
KL $\rightarrow$ MLE, cross-entropy $\rightarrow$ log-loss, forward/reverse KL,
maximum entropy principle, and mutual information.

KL = Maximum Likelihood

The single most important connection between
information theory and machine learning.

Minimizing KL = Maximizing Log-Likelihood

Data from true p , model family q_θ . Find θ making q_θ closest to p .

Step 1: Write out $D_{\text{KL}}(p\|q_\theta) = \sum_x p(x)[\log p(x) - \log q_\theta(x)]$.

Which of those two terms depends on θ ?

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$$\arg \min_{\theta} D_{\text{KL}}(p\|q_\theta) = \arg \max_{\theta} \sum_x p(x) \log q_\theta(x)$$

Step 3: Replace p by empirical $\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[x_i = x]$:

$$\arg \max_{\theta} \sum_x \hat{p}(x) \log q_\theta(x) = \arg \max_{\theta} \frac{1}{n} \sum_{i=1}^n \log q_\theta(x_i)$$

$$\arg \min_{\theta} D_{\text{KL}}(p\|q_\theta) = \arg \max_{\theta} \sum_{i=1}^n \log q_\theta(x_i) = \textbf{MLE!}$$

Cross-Entropy = MLE: A Concrete Example

Since $H(p||q_\theta) = H(p) + D_{\text{KL}}(p||q_\theta)$ and $H(p)$ is constant in θ : the same optimum solves all three.

Minimize KL

=

Minimize cross-entropy

=

Maximize log-likelihood

Numerical example. 3-class classifier, true class is class 1. Loss is $-\log_2 \pi_1$:

Predicted (π_1, π_2, π_3)	Loss (bits)	Interpretation
(0.7, 0.2, 0.1)	0.51	confident, correct
(0.4, 0.4, 0.2)	1.32	uncertain
(0.1, 0.1, 0.8)	3.32	wrong, but “not too sure”
(0.05, 0.05, 0.9)	4.32	confidently wrong!
(1.0, 0.0, 0.0)	0	confident, correct (saturates)

Cross-entropy is **harsh on confident wrongness** — that’s the whole point.
Halving the predicted probability *doubles* the loss.

Cross-Entropy Loss in Classification

Multi-class classification with g classes. For observation i :

- ▶ True label $y_i \in \{1, \dots, g\}$, one-hot encoded as $\mathbf{d}^{(i)} = (0, \dots, 1, \dots, 0)$
- ▶ Model outputs probability vector $\boldsymbol{\pi}(\mathbf{x}_i \mid \theta) = (\pi_1, \dots, \pi_g)$

The cross-entropy between one-hot $\mathbf{d}^{(i)}$ and model $\boldsymbol{\pi}$:

$$H(\mathbf{d}^{(i)} \parallel \boldsymbol{\pi}) = - \sum_{k=1}^g d_k^{(i)} \log \pi_k(\mathbf{x}_i \mid \theta) = - \log \pi_{y_i}(\mathbf{x}_i \mid \theta)$$

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Summing over all observations:

$$\mathcal{R}(\theta) = - \sum_{i=1}^n \log \pi_{y_i}(\mathbf{x}_i | \theta) \quad (= \text{negative log-likelihood} = \text{log-loss})$$

Cross-entropy loss IS log-loss IS negative log-likelihood.

Three names for the same thing. Softmax + cross-entropy = the standard recipe.

Binary Cross-Entropy = Bernoulli Log-Loss

Binary classification: $y \in \{0, 1\}$, model outputs $\pi(\mathbf{x}) = P(Y=1 \mid \mathbf{x})$.

$$L(y, \pi) = -y \log \pi(\mathbf{x}) - (1 - y) \log(1 - \pi(\mathbf{x}))$$

This is the cross-entropy between two Bernoulli distributions: $p = \text{Bern}(y)$ and $q = \text{Bern}(\pi(\mathbf{x}))$.

Connection to MLE / logistic regression: Logistic regression maximizes likelihood
 $\Leftrightarrow$ minimizes binary cross-entropy $\Leftrightarrow$ minimizes KL to the true conditional.
The cross-entropy loss in logistic regression **is** information-theoretic!

Entropy as Baseline Risk

What's the best a “dumb” model can do? Use class frequencies: $\pi_k = n_k/n$. Its average cross-entropy is just the class entropy:

$$\mathcal{R} = - \sum_k \pi_k \log \pi_k = H(\pi).$$

Entropy of the class distribution = risk of the best constant model. The **reduction** from $H(\pi)$ is what your model actually learns.

Class balance	Baseline entropy	Interpretation
(0.5, 0.5)	1 bit	Hard (max uncertainty)
(0.9, 0.1)	0.47 bits	Easy (already pretty sure)
(0.99, 0.01)	0.08 bits	Trivial (almost deterministic)

Imbalanced classes: predicting the majority class always gives 99% accuracy on (0.99, 0.01) data — but only matches the 0.08-bit baseline. **Use loss reduction, not accuracy.**

Forward vs Reverse KL

KL is not symmetric. The direction matters — a lot.

Two Directions, Two Behaviors

We want to approximate a complex distribution p with a simpler model q_ϕ .

Forward KL: $D_{\text{KL}}(p \| q_\phi)$

- ▶ Averages $\log \frac{p}{q_\phi}$ under p
- ▶ Penalizes $q_\phi \approx 0$ where $p > 0$
- ▶ q_ϕ must **cover all** of p 's mass
- ▶ Behavior: **mass-covering**
- ▶ Used in: supervised learning, MLE

Reverse KL: $D_{\text{KL}}(q_\phi \| p)$

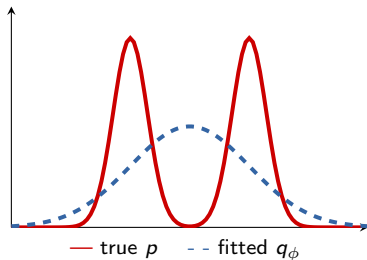
- ▶ Averages $\log \frac{q_\phi}{p}$ under q_ϕ
- ▶ Penalizes $q_\phi > 0$ where $p \approx 0$
- ▶ q_ϕ avoids regions where p is small
- ▶ Behavior: **mode-seeking**
- ▶ Used in: variational inference

Forward KL: "I'd rather be too wide than miss anything."

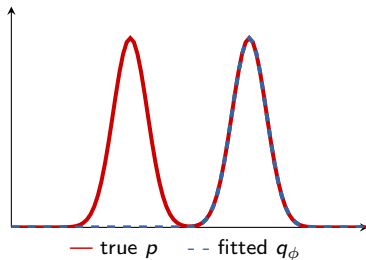
Reverse KL: "I'd rather be too narrow than put mass where p doesn't."

Forward vs Reverse KL: Fitting a Gaussian to a Bimodal

Forward KL: mass-covering



Reverse KL: mode-seeking



Forward KL produces a **wide** fit covering both modes (but too flat).
Reverse KL locks onto **one mode** (sharp but misses the other).

Why: forward KL averages under p (penalizes $q_\phi \approx 0$ where $p > 0$); reverse KL averages under q_ϕ (just avoids regions of low p).

Why Each Direction Is Used

	Forward KL: $D_{\text{KL}}(p \ q_\phi)$	Reverse KL: $D_{\text{KL}}(q_\phi \ p)$
Gradient	Requires samples from p	Requires samples from q_ϕ
Works when	p is the empirical distribution	p is unnormalized (up to Z)
Use case	Supervised learning (MLE)	Variational inference (VI)
Behavior	Mass-covering (inclusive)	Mode-seeking (exclusive)

Why VI uses reverse KL — the key trick:

In Bayesian inference the posterior $p(\theta \mid \mathbf{X}) = \frac{p(\theta, \mathbf{X})}{Z}$, where $Z = \int p(\theta, \mathbf{X}) d\theta$ is intractable.

But: $\nabla_\phi D_{\text{KL}}(q_\phi \| p) = \nabla_\phi \mathbb{E}_{q_\phi}[\log q_\phi(\theta) - \log p(\theta, \mathbf{X}) + \log Z]$, and $\nabla_\phi \log Z = 0$ since Z doesn't depend on ϕ .

The intractable normalizer drops out. This is the entire reason VI exists.

Maximum Entropy Principle

Given partial knowledge, choose the distribution that is **maximally uncertain** about everything else.

The Maximum Entropy Principle (Jaynes, 2003)

We know constraints (e.g., $\mathbb{E}[g_m(X)] = \alpha_m$ for $m = 1, \dots, M$). Many distributions satisfy them. Which to pick?

MaxEnt Principle: of all distributions satisfying the constraints, choose the one with **maximum entropy**.

Shore & Johnson, 1980: MaxEnt is the unique inference rule satisfying consistency under coarsening, independence, and continuity.

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Solution (Lagrangian):

$$p^*(x) = \frac{1}{Z} \exp\left(\sum_{m=1}^M \lambda_m g_m(x)\right) \quad Z = \sum_x \exp\left(\sum_m \lambda_m g_m(x)\right)$$

The MaxEnt distribution is always an **exponential family** (Gibbs distribution)!

This is no coincidence — exponential families **are** the MaxEnt distributions.

MaxEnt Recovers Familiar Distributions

Different constraints $\Rightarrow$ different MaxEnt distributions:

Constraint(s)	MaxEnt distribution	Connection
None (fixed support $\{1, \dots, g\}$)	Uniform	Max entropy = $\log g$
$\mathbb{E}[X] = \mu$ (on $\{1, \dots, g\}$)	Exponential/Gibbs	Biased die
$\mathbb{E}[X] = \mu, \text{Var}(X) = \sigma^2$	Gaussian $N(\mu, \sigma^2)$	Most common in ML!
$\mathbb{E}[X] = 1/\lambda$ (on $[0, \infty)$)	Exponential (λ)	Memoryless

Why the Gaussian is everywhere in ML: if you only know the mean and variance, the Gaussian is the **least informative** distribution compatible with that knowledge — any other choice smuggles in extra assumptions.

The CLT, information-theoretically: averaging more i.i.d. variables makes the (normalized) sum's entropy climb *monotonically* to the maximum — the Gaussian is the **entropy attractor of summation**. Keep convolving, and you converge to the bell curve.

Proof: Gaussian Maximizes Differential Entropy

Claim: Among all densities p on $\mathbb{R}$ with mean μ and variance σ^2 , the Gaussian $g = N(\mu, \sigma^2)$ maximizes differential entropy.

Proof (via Gibbs' inequality applied to differential entropy):

Let p be any density with the same mean and variance as g . Consider $D_{\text{KL}}(p\|g) \geq 0$:

$$0 \leq D_{\text{KL}}(p\|g) = \int p(x) \log_2 \frac{p(x)}{g(x)} dx = -h(p) - \int p(x) \log_2 g(x) dx.$$

Plug in $\log_2 g(x) = -\frac{1}{2} \log_2(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2 \ln 2}$ and take the integral. Since p has the *same* mean and variance as g , the integral of $-\log_2 g(x)$ under p equals its integral under g :

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$$-\int p(x) \log_2 g(x) dx = -\int g(x) \log_2 g(x) dx = h(g) = \frac{1}{2} \log_2(2\pi e \sigma^2).$$

Substituting back: $0 \leq -h(p) + h(g) \Rightarrow h(p) \leq h(g)$. ■

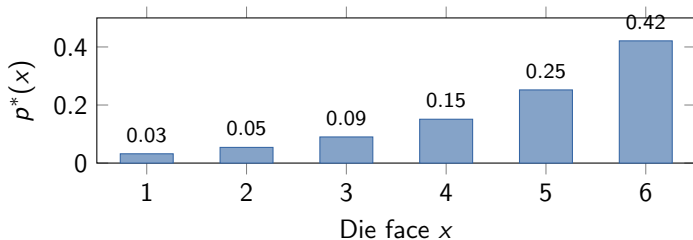
Same trick works for every MaxEnt distribution in the table above: pick the candidate distribution, write $D_{\text{KL}}(p\|\text{candidate}) \geq 0$, exploit matched moments. The exponential family is essentially the MaxEnt family.

MaxEnt Example: The Biased Die

A 6-sided die with $\mathbb{E}[X] = 4.8$ (heavier than the usual 3.5). What's the MaxEnt distribution?

$$p^*(x) = \frac{e^{\lambda x}}{Z(\lambda)}, \quad Z(\lambda) = \sum_{x=1}^6 e^{\lambda x}, \quad \sum_{x=1}^6 x p^*(x) = 4.8$$

Solving numerically: $\lambda \approx 0.514$.



Exponential tilt toward higher faces — the “least opinionated” way to have $\mathbb{E}[X] = 4.8$.

Sign of λ encodes the deviation: $\lambda = 0 \rightarrow$ uniform; $\lambda > 0 \rightarrow$ higher faces;
 $\lambda < 0 \rightarrow$ lower.

Mutual Information

How much does knowing X tell us about Y ?

A universal measure of dependence.

Joint and Conditional Entropy

Joint entropy: Total uncertainty in (X, Y) together:

$$H(X, Y) = - \sum_{x,y} p(x, y) \log p(x, y)$$

Conditional entropy: Remaining uncertainty in Y after observing X :

$$H(Y | X) = - \sum_{x,y} p(x, y) \log p(y | x) = \mathbb{E}_X[H(Y | X=x)]$$

Two natural questions:

- ▶ How does $H(X, Y)$ decompose? (Answer: the *chain rule*, next slide.)
- ▶ How much does Y *reduce* our uncertainty about X ? (Answer: *mutual information*, in 2 slides.)

Joint and conditional entropy will give us the two key identities of the rest of this lecture.

Chain Rule: The Workhorse Identity

The chain rule **factors** joint uncertainty into stages. Both directions are valid:

$$H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y).$$

Symmetry consequence: $H(X) - H(X|Y) = H(Y) - H(Y|X)$. (This will be *mutual information* in 2 slides.)

Worked example: two coin flips $X, Y \in \{0, 1\}$, both fair, with $Y = X \text{ XOR } Z$ where $Z \sim \text{Bern}(0.1)$ independent of X .

- $H(X) = 1$ (fair coin).
- $H(Y|X) = H(Z) = h(0.1) \approx 0.469$ (binary entropy of 0.1).
- $H(X, Y) = H(X) + H(Y|X) \approx 1.469$ bits.

Sanity check the other direction: Y marginally is also fair ($\Rightarrow H(Y) = 1$), and by symmetry $H(X|Y) \approx 0.469$. Same total. ✓

n -variable generalization: $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$.

This is exactly how autoregressive language models factor the joint probability of a sequence.

Mutual Information: Three Equivalent Definitions

Question: How much does knowing X **reduce** our uncertainty about Y ?

Form 1 — uncertainty reduction:

$$I(X; Y) = H(X) - H(X | Y).$$

Knowing Y shrinks our uncertainty about X by this much.

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Form 2 — by symmetry of the chain rule:

$$I(X; Y) = H(Y) - H(Y | X) = H(X) + H(Y) - H(X, Y).$$

MI is symmetric in X and Y — unlike KL.

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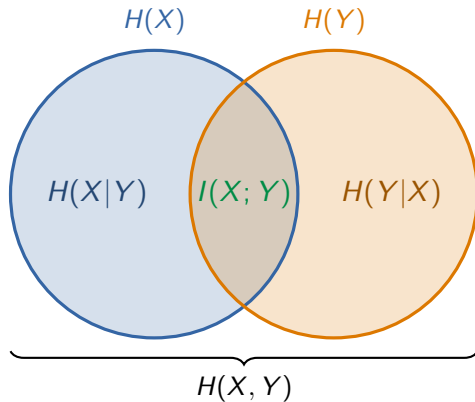
Form 3 — KL from independence:

$$I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y)) = \sum_{x, y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

MI = how far the joint is from the product of marginals.

$I(X; Y) = 0$ iff $X \perp\!\!\!\perp Y$. The larger I , the more X and Y “know” about each other.

The Information Diagram



$I(X; Y) = H(X) + H(Y) - H(X, Y)$ — the overlap = shared information.

Properties of Mutual Information

1. Non-negative: $I(X; Y) \geq 0$ (since $D_{\text{KL}} \geq 0$)

2. Zero iff independent: $I(X; Y) = 0 \Leftrightarrow X \perp\!\!\!\perp Y$

3. Symmetric: $I(X; Y) = I(Y; X)$ (unlike KL!)

4. Self-information: $I(X; X) = H(X)$

5. Bounded: $I(X; Y) \leq \min\{H(X), H(Y)\}$ (for discrete RVs)

6. Invariant: under invertible smooth transformations of X or Y

Conditioning reduces entropy (“information can’t hurt”):

$H(X | Y) \leq H(X)$, with equality iff $X \perp\!\!\!\perp Y$.

Proof: $0 \leq I(X; Y) = H(X) - H(X | Y)$.

Worked Example: Computing MI

Joint distribution $p(x, y)$:

$X \backslash Y$	y_1	y_2	y_3	y_4
x_1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
x_2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
x_3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
x_4	$\frac{1}{4}$	0	0	0

Marginals: $p_X = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $p_Y = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$.

Group the joint by value (read off the table):

$$\blacktriangleright 1 \times \frac{1}{4}, \quad 2 \times \frac{1}{8}, \quad 6 \times \frac{1}{16}, \quad 4 \times \frac{1}{32}$$

$$H(X) = \log_2 4 = 2 \text{ (uniform).}$$

$$H(Y) = \frac{1}{2}(1) + \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) = \frac{7}{4}.$$

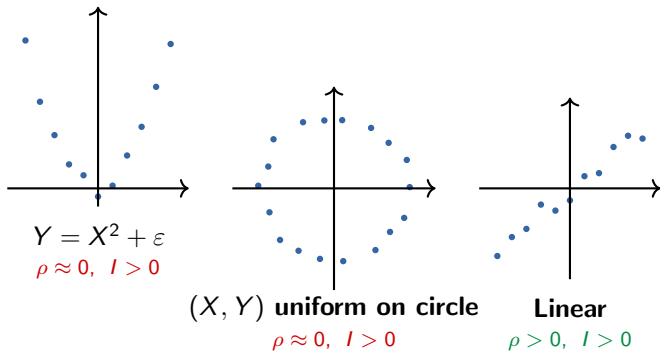
$$\begin{aligned} H(X, Y) &= \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) + 6 \cdot \frac{1}{16}(4) + 4 \cdot \frac{1}{32}(5) \\ &= \frac{4}{8} + \frac{6}{8} + \frac{12}{8} + \frac{5}{8} = \frac{27}{8}. \end{aligned}$$

Mutual information:

$$I(X; Y) = 2 + \frac{7}{4} - \frac{27}{8} = \boxed{\frac{3}{8} \text{ bits}}$$

MI vs Correlation: Why MI Is More General

Pearson correlation only measures **linear** dependence. MI captures **any** dependence.



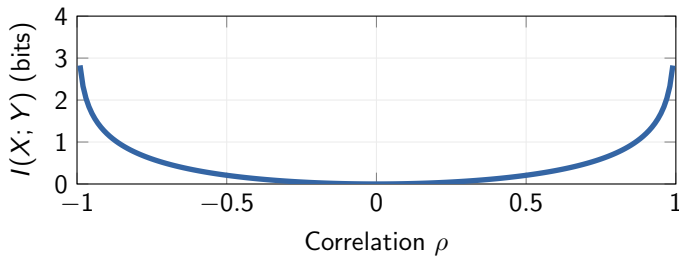
Correlation = 0 does NOT mean independence. MI detects **any** dependence — linear, nonlinear, or otherwise.

MI for Correlated Gaussians

For any $(X, Y) \sim N\left(\boldsymbol{\mu}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}\right)$:

$$I(X; Y) = -\frac{1}{2} \log_2(1 - \rho^2).$$

Why: $h(X) + h(Y) - h(X, Y) = \frac{1}{2} \log(2\pi e \sigma_X^2 \cdot 2\pi e \sigma_Y^2) - \frac{1}{2} \log((2\pi e)^2 \sigma_X^2 \sigma_Y^2 (1 - \rho^2))$. All but ρ cancels.

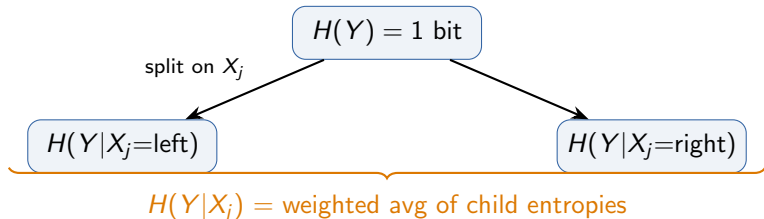


$\rho = 0 \Rightarrow I = 0$ (Gaussian: uncorrelated = independent). $|\rho| \rightarrow 1 \Rightarrow I \rightarrow \infty$.

MI in ML: Information Gain in Decision Trees

At each node of a decision tree, we pick the feature that **maximizes information gain**:

$$IG(Y; X_j) = H(Y) - H(Y | X_j) = I(Y; X_j)$$



$$\mathbf{IG} = H(Y) - H(Y|X_j) = I(Y; X_j)$$

Split on the feature with highest MI with the target.

Exactly the criterion in ID3 and C4.5. CART uses Gini by default but can use entropy. Feature selection: rank features by $I(X_j; Y)$, pick top ones.

The Information Theory Toolbox for ML

Concept	Formula	ML Role
Entropy $H(p)$	$-\sum p \log p$	Baseline risk, impurity
Cross-entropy $H(p\ q)$	$-\sum p \log q$	Loss function (log-loss)
KL divergence D_{KL}	$\sum p \log \frac{p}{q}$	MLE, variational inference
Mutual info $I(X; Y)$	$D_{\text{KL}}(p_{xy} \ p_x p_y)$	Feature selection, info gain
Forward KL	$D_{\text{KL}}(p \ q_\theta)$	Supervised learning
Reverse KL	$D_{\text{KL}}(q_\phi \ p)$	Variational inference
MaxEnt	$\arg \max H$ s.t. constraints	Gaussian, exp family

Key takeaway: Information theory is not just “theory” — it directly gives us the loss functions, optimization objectives, and model selection criteria we use every day in ML.

Questions?